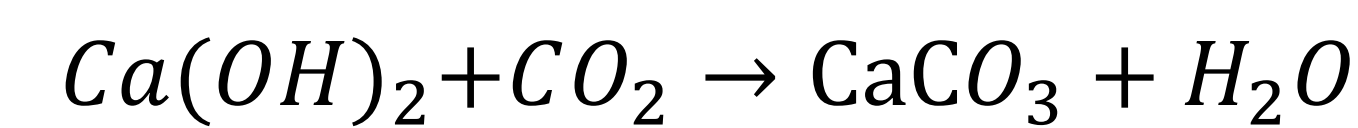
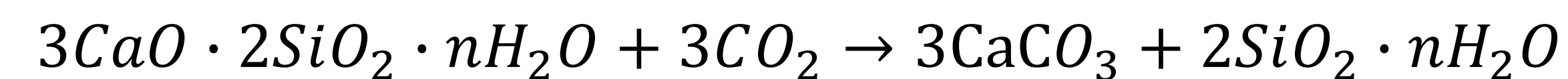
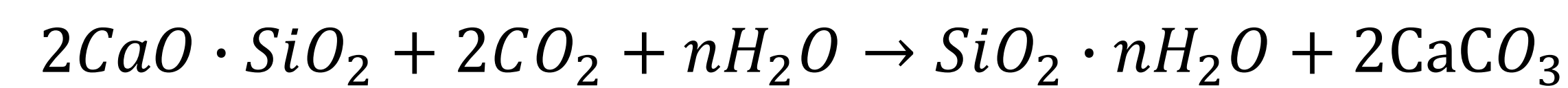
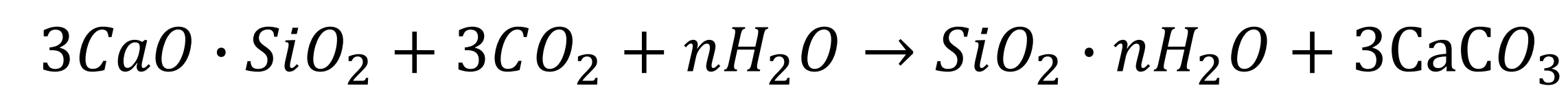


Is Accelerated Carbonation a Benefit or Burden to Cementitious Materials? A Look at Volume Stability

I Introduction

With climate change intensifying, carbon capture, utilization, and storage (CCUS) has become essential for environmental protection. Among CCUS methods, carbonation curing offers a promising path to carbon-neutral concrete by reacting CO_2 with cement clinker and hydration products, forming stable carbonates:



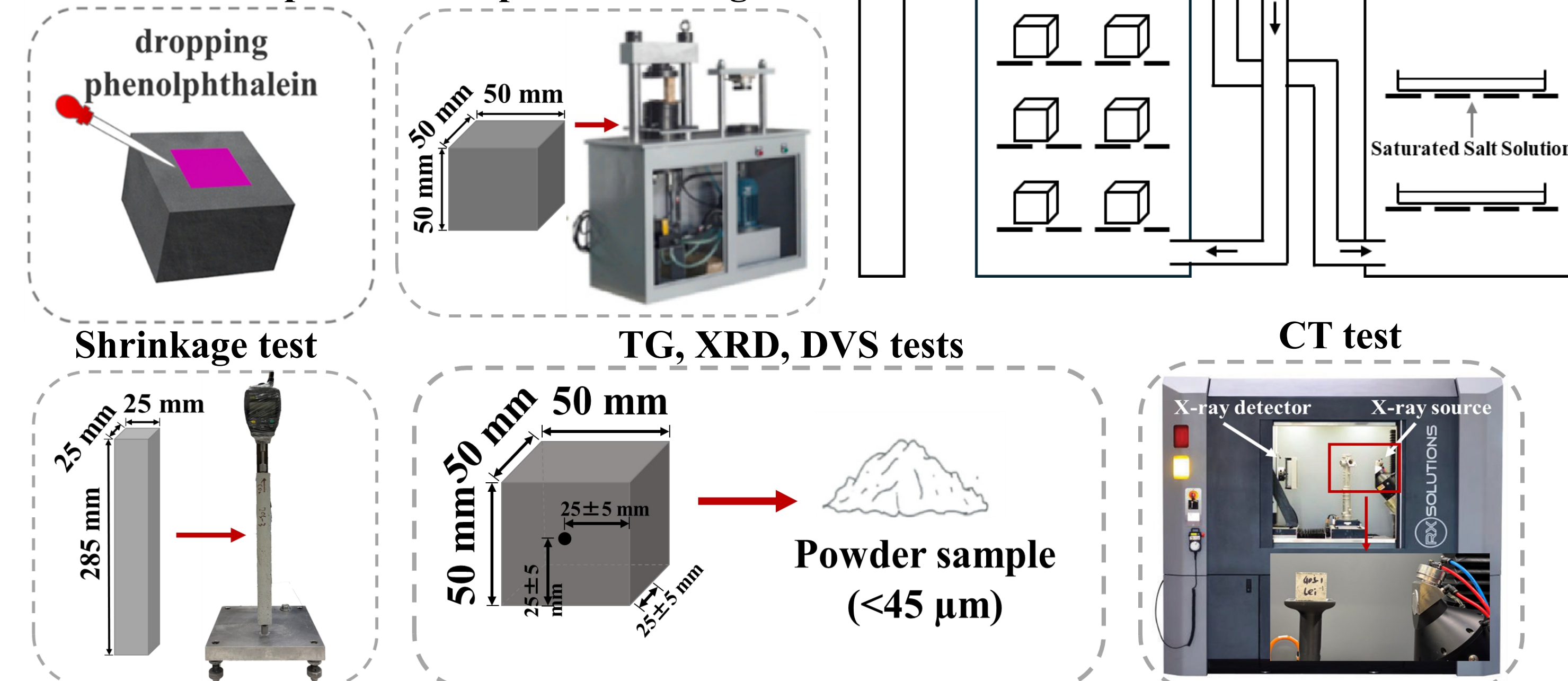
Challenge: maintain volume stability during accelerated carbonation curing while retaining mechanical properties and accelerating the carbonation process.

Motivation: investigate the impact of accelerated carbonation (3% CO_2) on the properties of cementitious materials using 0-80% GGBFS to keep volume stability and enhance sustainability.

II Methods

Carbonation depth, compressive strength, shrinkage were evaluated. Phase changes were analyzed by XRD and TGA, and pore structure was characterized by DVS and XCT.

Carbonation depth test Compressive strength test



3.1. Macroscopic properties: carbonation depth and carbonation rate

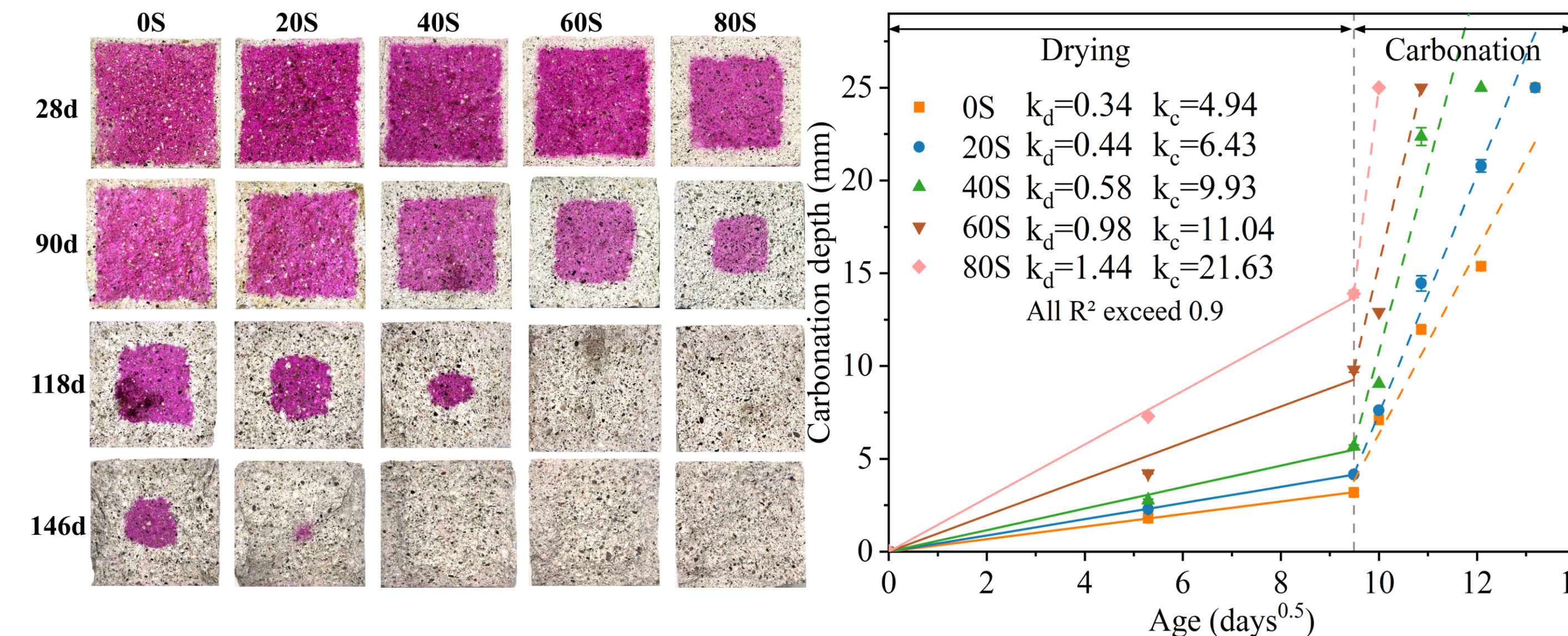


Fig. 1. Carbonation depth and carbonation rate of specimens exposed to carbonation conditions

- Incorporation of GGBFS leads to higher carbonation depth (d) and carbonation rate (k) ($d = k \cdot \sqrt{t}$).
- Carbonation depth and rate increase over time. Accelerated carbonation raises the carbonation rate by up to 15 times compared to that under dry conditions.

3.2. Macroscopic properties: compressive strength and shrinkage

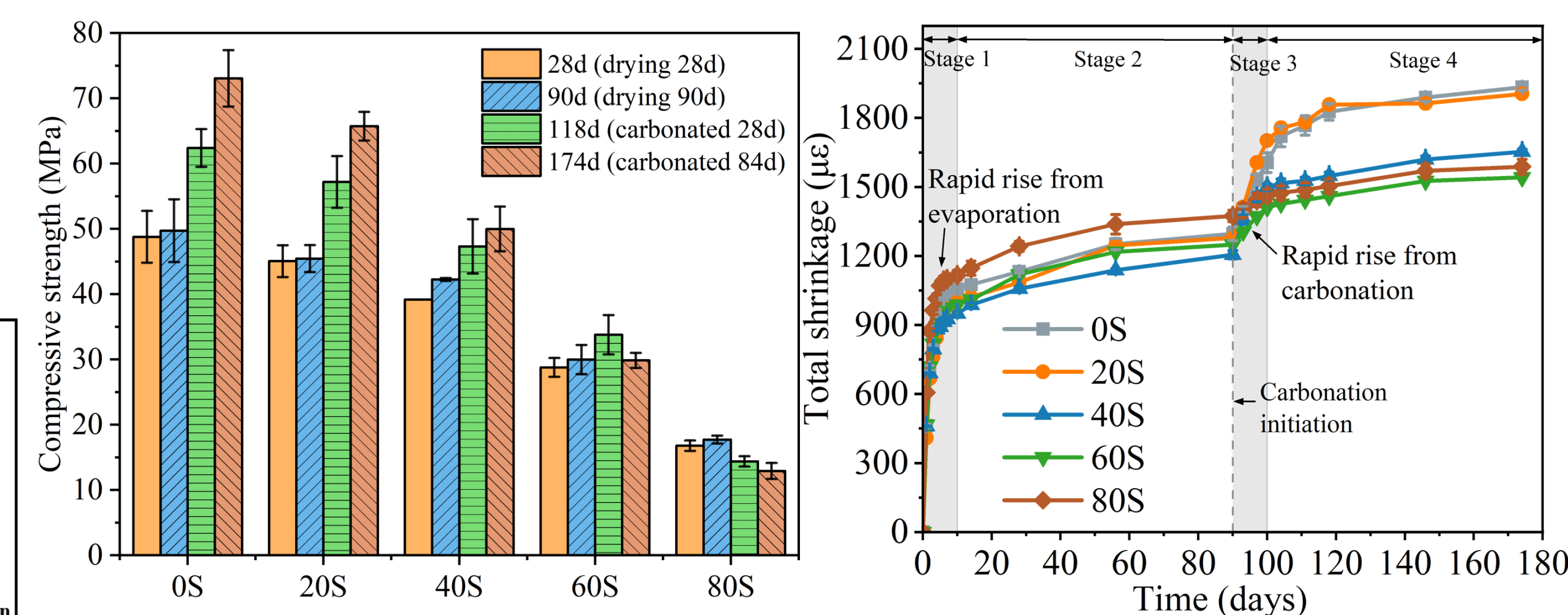


Fig. 2. Compressive strengths of specimens subjected carbonation curing

Fig. 3. Shrinkage of specimens subjected carbonation curing

- GGBFS-blended specimens show lower compressive strength under dry conditions compared to plain specimens, but this is compensated by accelerated carbonation.
- Carbonation increases shrinkage and leads to volume instability, but incorporating GGBFS effectively mitigates this effect.

III Results

3.3. Microscopic properties: TG and XRD

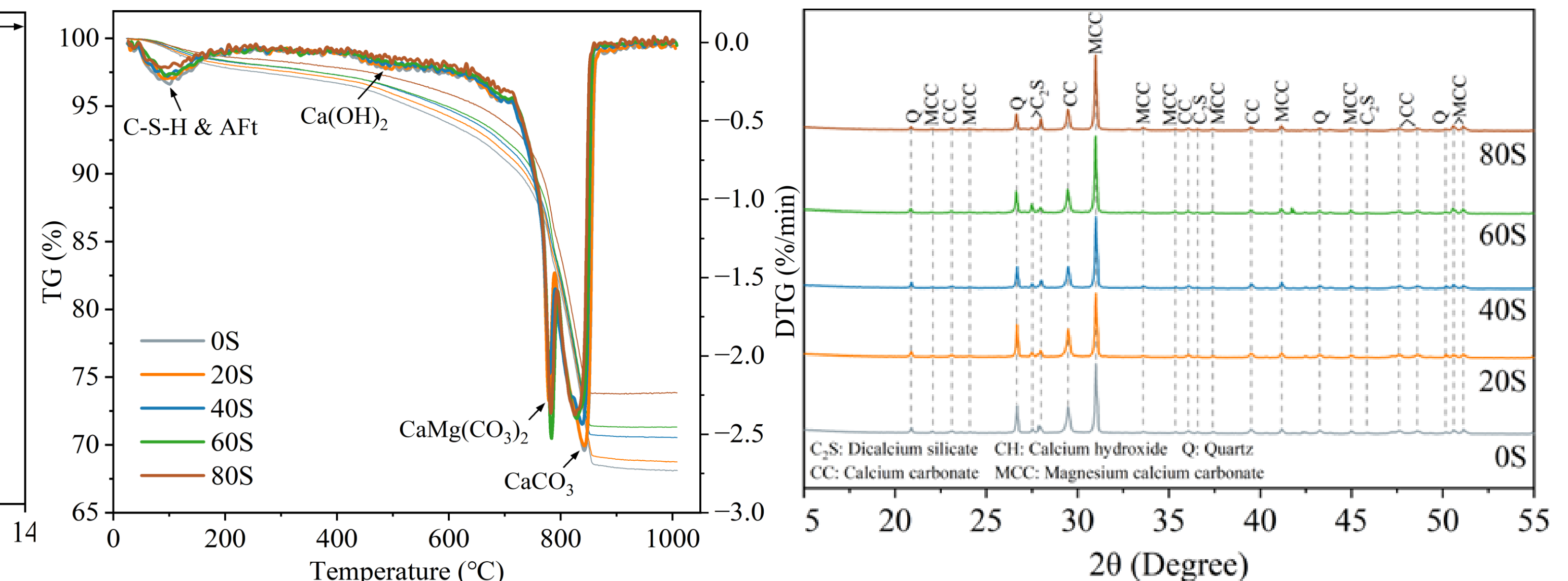


Fig. 4. TG and XRD analysis of GGBFS-blended specimens under carbonation conditions at 174 days

- Dolomite is found beneficial for strength, as its magnesium content promotes M-S-H gel formation, improving microstructure.

3.4. Microscopic properties: CT and DVS

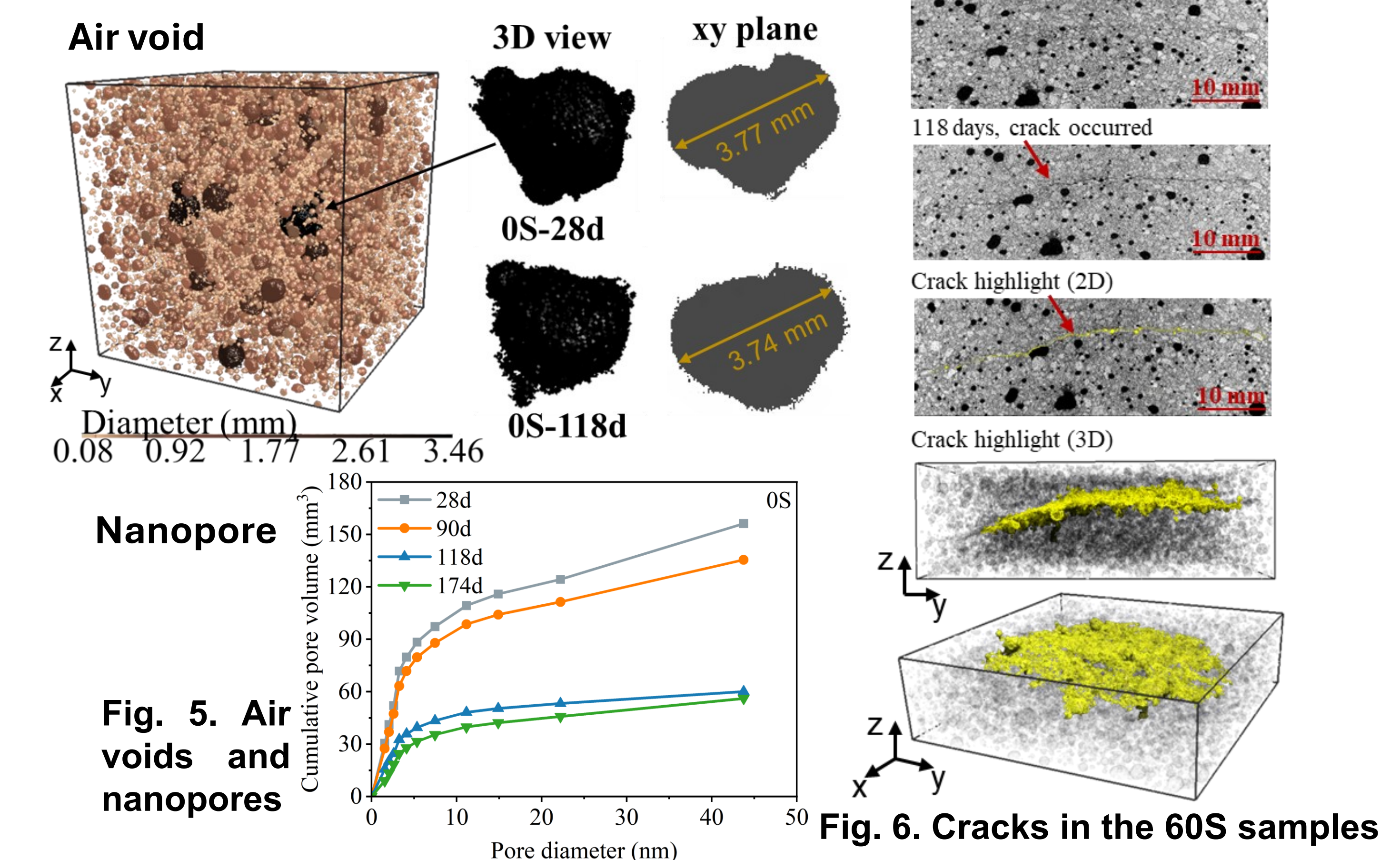


Fig. 5. Air voids and nanopores

Fig. 6. Cracks in the 60S samples

- Air void and nanopore refinement after carbonation increase compressive strength.
- Presence of cracks in the samples with 60% GGBFS leads to a reduction in compressive strength.

IV Conclusion

Benefit: accelerated carbonation enhances both compressive strength and carbonation rate.

Burden: shrinkage increases during the accelerated carbonation.

Solution: incorporation of GGBFS reduces carbonation shrinkage, contributing to more volume-stable cementitious materials.

References

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Acknowledgements

The authors gratefully acknowledge the support of the University of Toronto and the funding provided by CSBE.